

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering  
**ASSESSMENT TOOL 2 – CONCEPT MAPPING**

|                  |                                                                                                                                                                |               |                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA0601                                                                                                                                                        | Course Title: | Design and Analysis of Algorithms   |
| CO Assessed:     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | Assessment:   | Assessment Tool 2 – Concept Mapping |
| Weightage:       | 20% of CIA                                                                                                                                                     | Total Marks:  | 100 (2 Questions × 50 Marks)        |
| CO1 Statement:   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |               |                                     |
| Student Name:    | U. Prabhavathi                                                                                                                                                 | Reg. No.:     | 192472061                           |
| Section / Batch: | CSE-AJ                                                                                                                                                         | Date:         | 20/07/26                            |

**Instructions to Students**

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

| Question Type   | Focus Area                                                                                                                      | Questions | Marks            |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|-----------|------------------|
| Concept Mapping | Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method | Q1 – Q2   | Q1 –50<br>Q2 –50 |
| GRAND TOTAL:    |                                                                                                                                 | 100 Marks |                  |

**SECTION A – CONCEPT MAPPING**

48  
50 ✓

**Code of Conduct**

I, V. Prabhavathi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [Signature]

**RUBRICS FOR CALCULATION**

| Criteria                  | Weightage | Level 4 – Exemplary                                                         | Level 3 – Proficient                              | Level 2 – Developing                                        | Level 1 – Beginning                     |
|---------------------------|-----------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%       | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%       | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%       | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%       | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |
| Clarity                   | 10%       | Concept map is clear, neat, visually structured, and easy to interpret      | Generally clear with minor readability issues     | Clarity is reduced; difficult to interpret in parts         | Poorly presented and hard to understand |

**Marks Evaluation:**

| Question No.        | Concept Identification (25%) | Linkages (25%) | Organization (20%) | Integration of Literature (20%) | Clarity (10%) | Total Marks (Each – 50) |
|---------------------|------------------------------|----------------|--------------------|---------------------------------|---------------|-------------------------|
| Q1                  | 25                           | 25             | 20                 | 18                              | 10            | 48                      |
| Q2                  |                              |                |                    |                                 |               |                         |
| Overall Total (100) |                              |                |                    |                                 |               |                         |

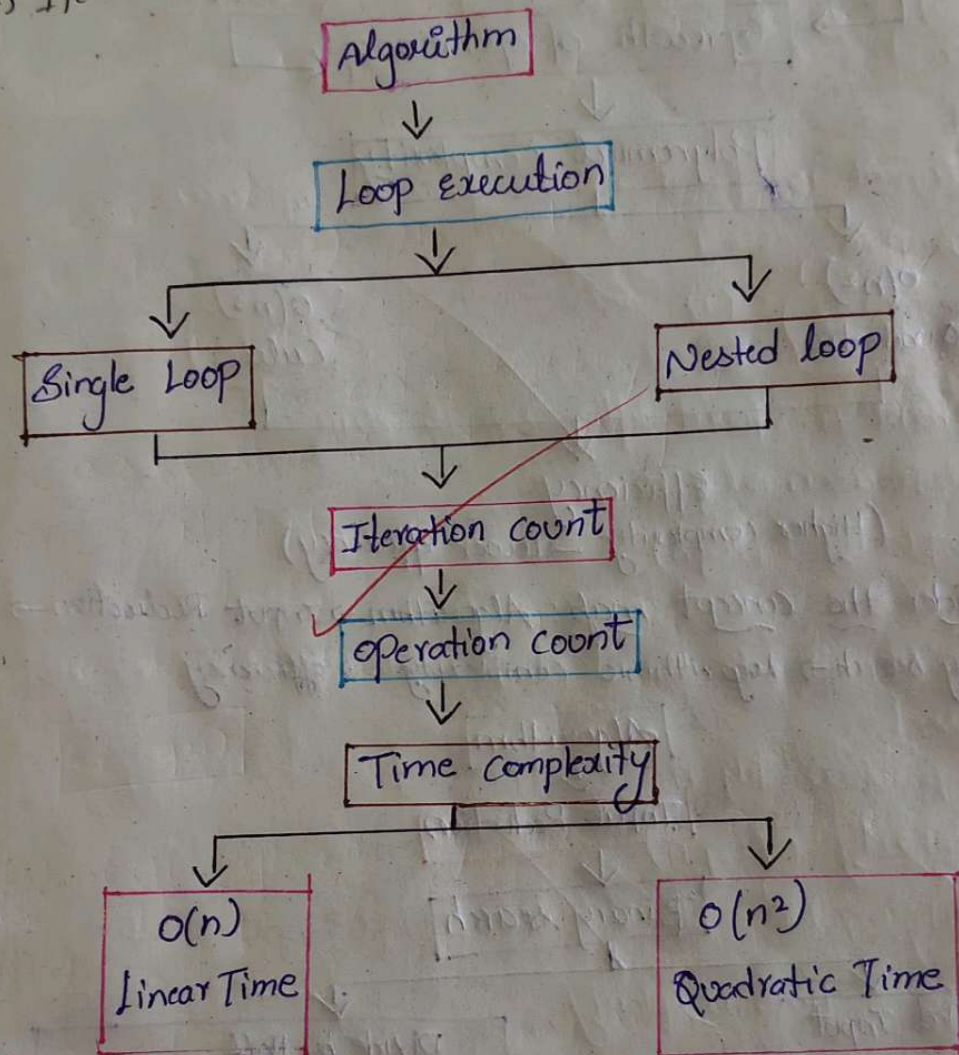
| Faculty In-charge             | Course Coordinator | Head of Department |
|-------------------------------|--------------------|--------------------|
| Signature: <u>[Signature]</u> | Signature: _____   | Signature: _____   |
| Date: <u>20/11/26</u>         | Date: _____        | Date: _____        |



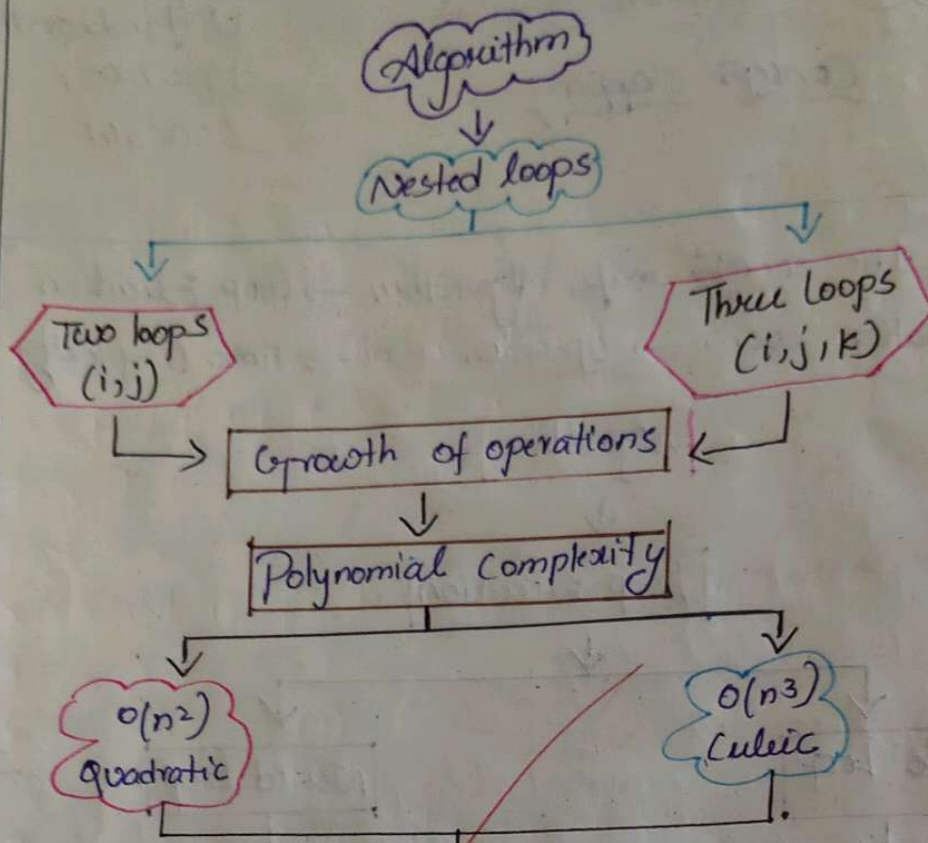
Col- Assessment Tool-2.  
Concept Mapping

U. Prabhavathi  
192472067  
(SA0601)

96 Consider the concept map: Algorithm  $\rightarrow$  Loop Execution  
 $\rightarrow$  Iteration count  $\rightarrow$  operation count  $\rightarrow$  Time complexity.



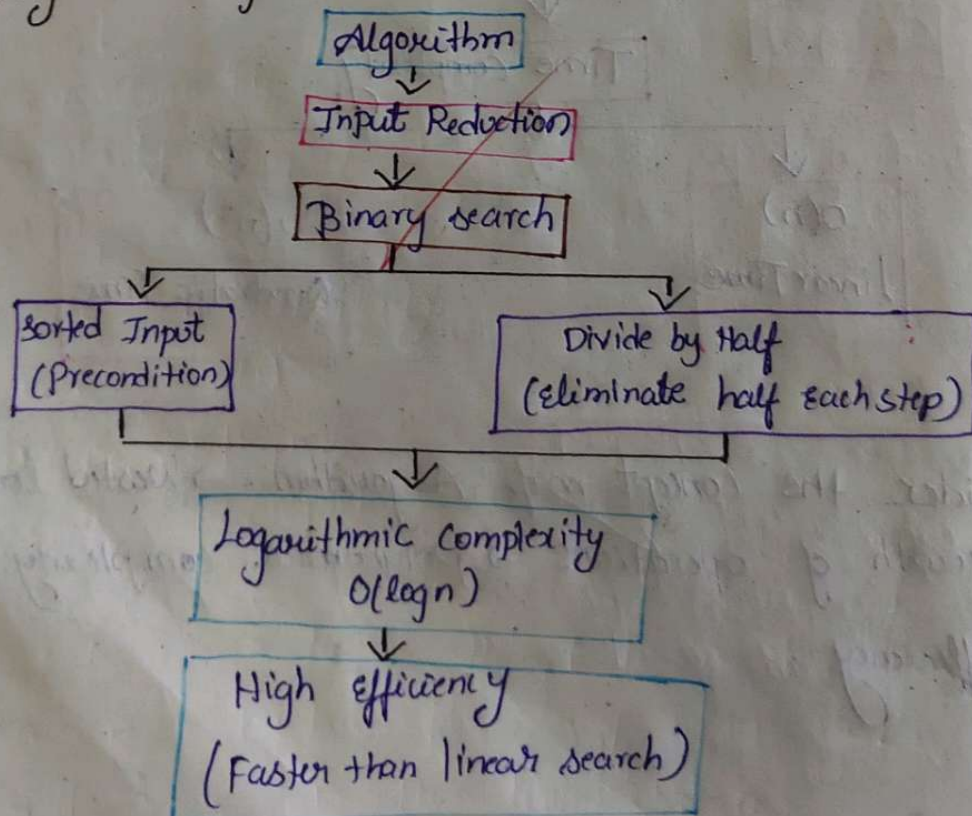
97 Consider the concept map: Algorithm  $\rightarrow$  Nested loops  
 $\rightarrow$  growth of operations  $\rightarrow$  Polynomial complexity  
 $\rightarrow$  efficiency.



Efficiency  
(Higher complexity → Lower efficiency)

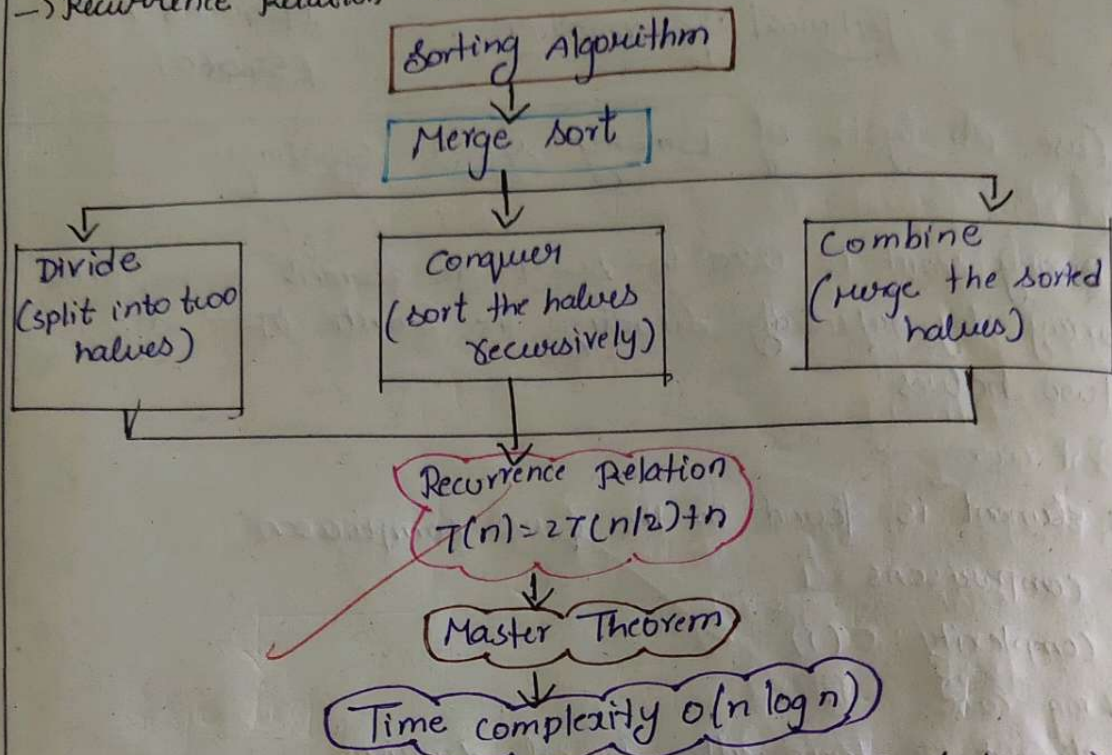
98

Consider the concept map: Algorithm → Input Reduction → Binary search → Logarithmic complexity → efficiency





99 Consider the concept map for sorting algorithm  $\rightarrow$  Merge sort  
 $\rightarrow$  Recurrence Relation  $\rightarrow$  Master Theorem  $\rightarrow O(n \log n)$



100 Consider the concept map :- Algorithm  $\rightarrow$  Brute Force  $\rightarrow$  High complexity  $\rightarrow$  optimization Techniques  $\rightarrow$  efficient Algorithm

